

1. An overview of the deliverables (state the task/project assigned to you, explain what you did and refer to outputs; you can mention challenges and barriers/limitations and highlight your achievements). If you have something along these lines written for the host, please use that to avoid duplicating the effort.

This spring and summer, I worked on the TUSAIL Experimental Measurements Database project with Professor Jin Ooi and Dr John Morrissey from the School of Engineering. [Training in Upscaling particle Systems: Advancing Industry across Length-scales | TUSAIL](#) is an EU Horizon 2020 grant funded project about training early career researchers in engineering and sharing simulation tools and methodologies with industry partners. My internship was on a part of this larger project, focusing on aggregating metadata about experiments with engineering materials – the TUSAIL Experimental Measurements Database or “EMDB” for short.

Before I joined the team, Dr Morrissey had already created a prototype EMDb website, available here: [TUSAIL EM Db \(jpmmor.eu.pythonanywhere.com\)](http://tusail-em-db.jpmmor.eu.pythonanywhere.com). This prototype was mostly complete from an end user’s perspective, but it was only suitable as a demo, running with an in-memory SQLite database backend which does not scale to support high internet traffic nor allows changes to be made easily after deployment. I spent the first few weeks researching different backend options until we decided to settle on PostgreSQL with part of the data stored in SQL columns and part in JSON blobs. Mixing the two approaches made development more complex but will hopefully allow the database to both perform well under high demand and make the schema relatively easy to change after deployment. I used SQL Alchemy to write the Object-Relational Model for PostgreSQL tables, which, in theory, would make switching to another SQL database like MariaDB trivial if required. The rest of my project was re-implementing the prototype website with the new backend and adding a few new features.

Most results are difficult to showcase with screenshots because the user interface looks the same, it’s how data is delivered to the interface that was completely overhauled. However, some of the new features are visible, for example, I extended the experiment schema to allow recording experiments with more than one material and/or equipment and updated the user interface to surface this information:

Before:

Home / Experiments Database

Experiments Database

The searchable experimental records database is shown below. New experiments can be added to the database.

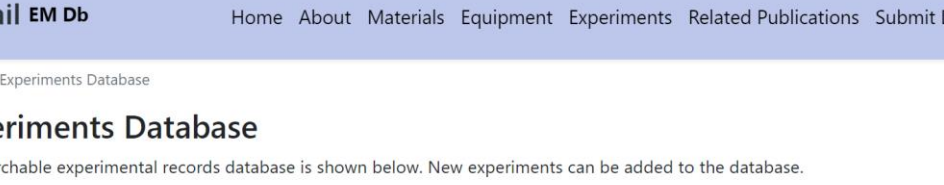
[View Material Metadata](#)
[View Equipment Metadata](#)
[Submit New Experimental Record](#)

Show entries Search:

ID	Date	User	Material	Equipment	Location	Category	Phase	Purpose	Quality	Dataset
1	2012-06-12	J.P. Morrissey	Limestone	Edinburgh Powder Tester	University of Edinburgh, Edinburgh, UK	Uniaxial Test	Dry	Calibration	3.7	gitlab.com/jpmmor

After:

After:



tusail EM Db

Home About Materials Equipment Experiments Related Publications Submit Data ▾ Contact

[Home](#) / Experiments Database

Experiments Database

The searchable experimental records database is shown below. New experiments can be added to the database.

[View Material Metadata](#) [View Equipment Metadata](#) [Submit New Experimental Record](#)

Show entries Search:

ID	Operator	Test Date	Materials and Equipment	Purpose	Dataset	Licence
1	Vid	2000-12-20	Test Material with Edinburgh pilot shear tester Test Material with Edinburgh big shear box	Verification	fake	ignored
2	Vid	2000-04-20	Another Material with Edinburgh huge rotating drum tester	Calibration	classified	ignored

Another new feature was making the units of measurements explicit and allowing users to convert between units on the website:

Before:

Before:															
<table><thead><tr><th colspan="2">Physical Properties</th></tr></thead><tbody><tr><td>True Density:</td><td>2800.0</td></tr><tr><td>Envelope Density:</td><td>2650.0</td></tr><tr><td>Specific Gravity:</td><td></td></tr><tr><td>Particle Porosity:</td><td></td></tr></tbody></table>		Physical Properties		True Density:	2800.0	Envelope Density:	2650.0	Specific Gravity:		Particle Porosity:					
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Hardness:	3.0														
Toughness:															
Ductility:															
After:															

After:

physical_properties			mechanical_properties		
colour	N/A		shear_modulus	N/A	Pa
coating_material	N/A		youngs_modulus	N/A	Pa
refractive_index	N/A	-	poissons_ratio	N/A	None
true_density	N/A	kg/m^3	mineral_hardness	N/A	mohs
envelope_density	N/A	kg/m^3	hardness	N/A	N/mm^2
specific_gravity	N/A	None	toughness	N/A	Pa
particle_porosity	N/A	None	ductility	N/A	None

Throughout the project, we had to make lots of design decisions. Frequent communication with the team (weekly meetings and/or catch-up emails) helped me keep making steady progress without getting blocked.

With a broad scope of initial tasks, there are still plenty of unfinished things left to do in the project. I didn't have time to polish the user interface layout, test changes extensively, nor implement all the desired new features. At the project midpoint and in the second last week, we discussed priorities with the team, and I focused my time on the bits that would be most challenging for them to complete by themselves after the handover. I made comprehensive notes throughout the project, including a handover summary at the end, including what I did, what is still left to do, and some tips and things to watch out for when developing the EMDB further.

At some points in the project, I was surprised to see that several popular Python libraries I used had bugs and lacked features. I submitted a couple of open-source contributions to the library codebases (e.g., [Handle Enum columns backed by Python Enums by Vidminas · Pull Request #164 · kvesteri/wtforms-alchemy \(github.com\)](#) and [unevaluatedProperties error gets raised before more relevant validation errors · Issue #1132 · python-jsonschema/jsonschema \(github.com\)](#)). This made some parts of the project significantly more difficult than I had anticipated, and I ended up achieving less than I initially aimed to do. Discussing priorities with the team at least meant that the most important features were covered.

2. An overview of your experience working as a Digital Research Ambassador Intern. (How useful was the internship to you? Think about the experience you gained, the skills you developed and the connections you made during the internship. Feel free to provide genuine feedback and mention any aspects of the internship you consider less beneficial or make suggestions of what should be done better next year). If you want to write a blog style text on this, you are more than welcome.

During this internship, I worked on a wide range of tasks – from design and accessibility considerations in the user interface to researching pros/cons of different databases to re-implementing backend services for an existing application to contributing to open-source libraries. Although I had done all these things in my work before, I had only superficial prior experience with databases, and learned a lot through the initial research phase. Working with JSON Schema during the implementation phase was an interesting follow-up to an internship I undertook with Esri Inc. in

2018, where I developed a YAML language server that also used JSON schema – back then, I only needed to learn the basics, and knew enough to propose this as a viable option for the EMDb, but I ended up learning it a lot more thoroughly in this project. I think this experience will be useful going forward, as it makes me more qualified as a full-stack developer and is indirectly relevant for my PhD work, because I will need to develop several prototype applications of search for educators.

Besides technical skills, I enjoyed the opportunity to network outside of the School of Informatics. The team I worked with were friendly and helpful, and I think I might consider working with them again in the future. Although it is past the internship contract date, we have planned a final meeting in August, when the host team are back from annual leave – it will be nice to meet in-person to wrap up.

My mentor within Digital Research Services, André, helped connect us for the first meeting and then joined in occasionally in the weekly ambassador catch-up meetings. I think it was good to have a point of contact in case there were any issues – but in my case there weren't any, and I was happy to take the lead in organising meetings and monitoring my own progress, so we didn't interact with André much.

It was good to meet the rest of the Digital Research Services team and other interns regularly. Because our projects had little overlap, without these meetings, I would have been completely disconnected from the rest of the work going on in Digital Research Services.

The DRS Ambassador scheme seemed to be aimed mainly at those who had never done an internship before; this was noticeable in parts of the induction and some team communications. Having had several previous internships and part-time jobs, this wasn't very relevant for me, and I thought it seemed unusual at postgraduate level. But from a show of hands at the induction meeting, I could see that it was relevant for many of the other interns. In my own project, I had the freedom to use a wide variety of skills and stretch my abilities, so I think it was a good overall balance for the mixed audience of ambassadors this year.

Thinking right back to the beginning, I felt the interview stage was strange. The projects were a secret, and the interview was so short that I didn't have time to ask much about who the team were or what to expect. Unitemps staff asked me to sign a contract before I knew what I was signing up for, which was a little uncomfortable, because it left a lot up to chance. I expected to work on user studies and human-computer interaction research but got a project on full-stack development instead. In the end, this was a pleasant surprise, and I was very happy with my assigned project, but it could have gone the other way too. I would suggest organising a second chat after initial interviews with all candidates to discuss projects and meet the host teams before final confirmations.

I'd also be keen for Unitemps to drop the requirement for 2 references before application. Even though I had several past employers to pick from, I still had trouble sourcing references, and I imagine it is even more difficult for those who don't have previous work experience, which defeats the point of the internship.

The timing of the internship was quite good – it had little overlap with the academic year. One of the reasons why I applied for the role is because I work as a teaching assistant throughout most of the year, but usually have less work in May–August, when teaching is on pause.

Working up to 9 hours per week fits the postgraduate student regulations, which stipulate that students should not work more than an average of 9 hours per week at university over a full year. In practice, because teaching assistant work has variable load during the semester, I tend to focus more

on my studies during the academic year, and more on work experience in the summer. This year, I worked on another internship alongside the DRS Ambassadors scheme to make full use of my availability. Although I personally prefer full-time work during the summer, I can see how some people might prefer 9h/week part-time work.

I didn't interact much with the ISG Student Worker Forum, but it was nice to know that the opportunity is there. It was also good to join the IS All Staff meeting in June, where I made some useful contacts in preparation for the Learning and Teaching Conference.

Overall, I think the DRS Ambassadors scheme was very well organised and I would recommend it for those looking to build up some work experience. It requires applicants to be flexible about what they do, and I would not recommend it for those who have a fixed idea of what they want to work on. It's a good way to meet people from other parts of the university, and therefore great for those in earlier stages of their studies or those intending to stay at the university after studies.

3. Optional: First draft of slides for a short presentation on your project, as will be required for the Ambassador Event in October. No need to provide final slides, just drafts.